

Have a small
whiteboard/pen to use



Molar Mass



Molar Mass (**M** or sometimes **MM**) is the mass in grams of one mole of a substance and has units of **g/mol**.

The periodic table may be used to calculate the Molar Mass of any compound as there is a direct conversion between the atomic mass unit (u) and the Molar Mass (M).

For example one atom of H has an atomic weight of **1.008 u**, while 1 mole of H has a Molar Mass of **1.008 g/mol**.

Converting to Molar Mass

- Boron has an atomic weight of _____ u, 1 mole of B has a Molar Mass of _____ g/mol
- Calcium has an atomic weight of _____ u, 1 mole of Calcium has a Molar Mass of _____ g/mol
- Hydrogen has an atomic weight of _____ u, 1 mole of Hydrogen has a Molar Mass of _____ g/mol
- Neodymium has an atomic weight of _____ u, 1 mole of Neodymium has a Molar Mass of _____ g/mol

Converting to Molar Mass

- Boron has an atomic weight of 10.81 u, 1 mole of B has a Molar Mass of 10.81 g/mol
- Calcium has an atomic weight of 40.08 u, 1 mole of Calcium has a Molar Mass of 40.08 g/mol
- Hydrogen has an atomic weight of 1.008 u, 1 mole of Hydrogen has a Molar Mass of 1.008 g/mol
- Neodymium has an atomic weight of 144.24 u, 1 mole of Neodymium has a Molar Mass of 144.24 g/mol

Converting to Molar Mass

The relationship between moles, Molar Mass and mass of a compound is:

$$\text{Amount in moles} = \frac{\text{mass (g)}}{\text{Molar Mass (g/mol)}}$$

This is often shortened to: $n = \frac{m}{MM}$

Re-Arranging the Molar Mass Equation

Make sure you understand this basic equation, then you can rearrange for others:

$$n = \frac{m}{MM}$$

To solve for mass, multiply both sides by MM and cancel out on the right side:

$$MM \times n = \frac{m}{\cancel{MM}} \times \cancel{MM}$$

Leaving you with: $m = n \times MM$

NOTES: you do NOT have to show the rearranging on assessments and you can rearrange before or after putting in numbers although BEFORE IS RECOMMENDED

Try showing the
(detailed) steps to
rearrange to solve for
MM.

To use this equation for
moles we need to
calculate molar mass
of compounds as well
as elements



Law of Definite Proportions

-Definition-

A specific compound always contains the same elements in definite proportions by mass.

-Layman's Terms-

Any molecule of one type of compound will always have the same subscripts.

Example: All Carbon Dioxide molecules have the formula CO_2

Calculating Molar Mass

What is the Molar Mass of Potassium bisulfate, KHSO_4 ?

Step 1) Determine which atoms are present and how many of each there are

1 X K
1 X H
1 X S
4 X O

If you feel confident,
complete the chart on
your note

NOTE: can also be named potassium hydrogen sulfate

Calculating Molar Mass

What is the Molar Mass of KHSO_4 ?

Step 1) Determine which atoms are present and how many of each there are.

Step 2) Substitute the atomic weight from your periodic table in for each atom.

$$1 \times \text{K} = 1 \times 39.1 \text{ g/mol}$$

$$1 \times \text{H} = 1 \times 1.01 \text{ g/mol}$$

$$1 \times \text{S} = 1 \times 32.1 \text{ g/mol}$$

$$4 \times \text{O} = 4 \times 16.0 \text{ g/mol}$$

Calculating Molar Mass

What is the Molar Mass of KHSO_4 ?

Step 1) Determine which atoms are present and how many of each there are.

Step 2) Substitute the atomic weight from your periodic table in for each atom.

Step 3) Perform any required multiplication and add all of the MM's up.

$$1 \times 39.1 \text{ g/mol} = 39.1 \text{ g/mol}$$

$$1 \times 1.01 \text{ g/mol} = 1.01 \text{ g/mol}$$

$$1 \times 32.1 \text{ g/mol} = 32.1 \text{ g/mol}$$

$$4 \times 16.0 \text{ g/mol} = 64.0 \text{ g/mol}$$

$$136.2 \text{ g/mol}$$

Using the equations

How many moles of KHSO_4 are in 250g? Solve on your whiteboards.

How many moles of H_2O are in 18.02 g? Solve on your whiteboards.

A stylized, layered mountain range graphic in shades of teal and blue, located in the bottom right corner of the slide.